Adaptive Thresholding for Robust Iterative Image Reconstruction from Limited Views Projection Data

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***Abstract*—X-ray computed tomography (CT) is generally obtained through a reconstruction of object attenuation function from its projection data acquired through many angle views. However, in some cases the number of projection views is limited to only a small number that is theoretically not enough for stable reconstruction. The theory of compressed sensing (CS) introduce a new framework for solving this problem. In this paper, we present a simple iterative thresholding image reconstruction algorithm that include a minimization of cost function that include Log-likelihood and ℓ_1 norm distance to a reference image. The resulting thresholding function in the reconstruction algorithm is approximated with adaptive function that can preserve low-contrast regions from incorrect thresholding. Moreover, we present a simple approach to handle registration error between the reconstructed image and the reference. This approach is based on using dynamic reference image that generated from previous estimation of the reference image and the current image estimate. The proposed method proved to be robust in image reconstruction from small number of projection views through a simulation study.**

I. Introduction

Computed Tomography (CT) imaging is generally obtained through a reconstruction of object attenuation function from its projection data acquired through many angle views. The sampling rate and acquisition orbit of the scanner are important factors to determine the exactness of the image reconstruction problem. This work investigates the problem of image reconstruction from a small number of projection views. It is well-known that the reduction of data acquisition sampling rate in the angular domain appears in the reconstructed images as streak artifacts. However, it becomes important to investigate stable reconstruction from a small number of projections in many medical imaging and industrial applications. The recently emerged theory of compressed sensing (CS) provides a new framework for this problem [1], [2]. Several approaches were proposed to solve this problem based on the theory of CS (e.g. [3]–[5]).

We consider that a reference image of the scanned object is *a priori* known. The cost function for image reconstruction is designed to include the *a priori* known information. In this study, we extending our previous work in [5] with an adaptive soft-thresholding reconstruction algorithm from limited projection data. The thresholding function is approximated with a corresponding convex function. Moreover, a better distance function is identified using global pixel information to control prior uncertainty when reference image is used.

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II. Proposed Method

A. Image reconstruction framework

We consider the following imaging model:

$$y_i \approx \text{Poisson}\ [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle)] \tag{1}$$

where $\vec{x} = (x_1, \ldots, x_n)^\top$ is the image vector, $\vec{y} = (y_1, \ldots, y_m)^\top$ be the measured counts, $\vec{b} = (b_1, \ldots, b_m)$ is the blank scan, $A = \{a_{ij}\}$ is the $m \times n$ system matrix and $\langle ., . \rangle$ is the regular inner product. The cost function for image reconstruction is defined as:

$$f(\vec{x}) = L(\vec{x}) + \gamma D_{\vec{u}}(\vec{x}), \tag{2}$$

where $L(\vec{x})$ is the log-likelihood function defined as:

$$L(\vec{x}) = \sum_{i=1}^{m} \left[b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle) + y_i \langle \vec{a_i}, \vec{x} \rangle \right], \tag{3}$$

γ is a hyperparameter, $\vec{u} = (u_1, \ldots, u_n)$ is the reference image and $D_{\vec{u}}(\vec{x})$ is a distance function between the image $\vec{x}$ and the reference image $\vec{u}$ and is defined as:

$$D_{\vec{u}}(\vec{x}) = \sum_{j=1}^{n} d(x_j, u_j), \qquad d(a, b) = |a - b|. \tag{4}$$

The cost function in Eq. (2) is designed to search for the feasible solution that fit with the data using the log-likelihood function and at the same time close to the reference image in sense of ℓ_1 norm measurements.

The iterative reconstruction algorithm is derived using the majorization-minimization strategy [4], [6] and is summarized as:

1 Initialization $x^{(0)} = \epsilon$, with $\epsilon > 0$, $k \leftarrow 0$
repeat
  for $j = 1 : n$ **do**
2 $p_j = x_j^k + x_j^k \frac{\sum_i a_{ij}\left[b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle) - y_i\right]}{\sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)}$
  $t_j = \gamma x_j^k / \sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)$
3 $x_j^{k+1} = Th(p_j, u_j, t_j)$
  end
  $k \leftarrow k + 1$
until *stopping criterion*;

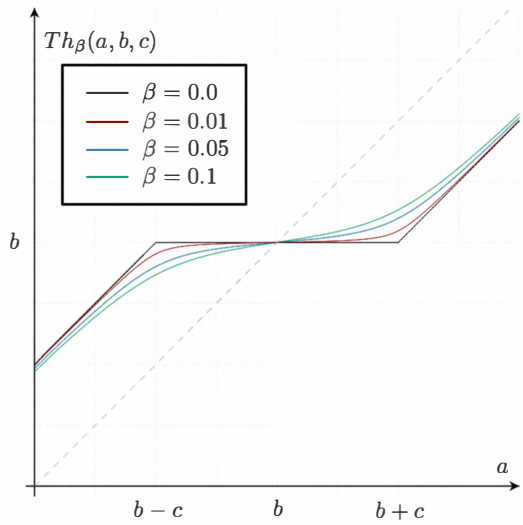

Fig. 1. Soft-thresholding function in Eq. (6) with different values of β.

where the thresholding function Th is defined as:

$$Th(a,b,c) = \begin{cases} a-c & (a > b+c) \\ a+c & (a < b-c) \\ b & (otherwise) \end{cases} . \quad (5)$$

In practical implementation, the soft-thresholding functions in Eq. (5), trims the pixel value p_j to the corresponding pixel value in the reference image u_j when it is relatively close. Otherwise, p_j is shifted *softly* towards the reference pixel intensity value [5].

B. Adaptive thresholding

The soft-thresholding function in Eq. (5) is non-convex. Due to this, incorrect thresholding may occur especially in pixels close to the boundaries. We generalize the soft-thresholding function by adding a convexity parameter β so that Eq. (5) is re-written as:

$$Th(a,b,c) = a + \frac{1}{2}\sqrt{(a-(b+c))^2+\beta} - \frac{1}{2}\sqrt{(a-(b-c))^2+\beta}, \quad (6)$$

where β is a small value that represents the convexity strength. When $\beta = 0$, Eqs. (5) and (6) become identicals as shown in Fig. 1.

C. Control priors uncertainty

One major drawback of practical implementation of reconstruction methods, that use a reference image similar to the one we propose here, is the consideration of the registration errors. Image shift or rotation of the registered reference image may reduce the image quality due to incorrect value of the corresponding reference pixel. In this regard, we propose a new global comparison strategy such that each image pixel x_j is associated to a corresponding window of a dynamic reference image instead of the comparison to a single pixel u_j in the reference image. The dynamic reference image is

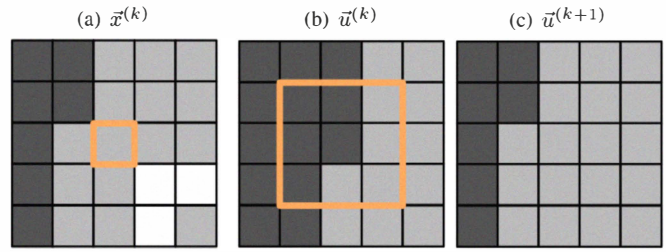

Fig. 2. Dynamic reference image, (a) is the current image estimate, (b) is the current reference image. The reference image for next iteration is computed using 3×3 window as in (c).

computed from the original reference and the current estimate of the reconstructed image such that:

$$u_j^{(k+1)} = \min_{u_i^{(k)}}\{d(x_j^{(k)}, u_i^{(k)})\}, i \in C \quad (7)$$

where C is a small window such as 3×3 pixels. In other words, each pixel of the reference image at the iteration $k+1$ is selected as one of the neighbor pixels in the reference image at iteration k, which is the closest to the corresponding pixel in the image estimate at iteration k. A simple example is shown in Fig. 2.

III. Experimental Results

We implemented a simulation study using the standard head Shepp-Logan phantom with image size 512×512 pixels and projection data measured using parallel-beam geometry using 512 (bins) and 32, 24, 20, and 16 projection angels over 180°. The reference image used is a simple uniform image shown in Fig. (3)(b). The proposed iterative reconstruction algorithm with the original thresholding function in Eq. (5) and the new one in Eq. (6) were implemented for 500 iterations. Reconstructed images and FBP reconstruction are shown in Fig. 3. The image reconstructed using the standard thresholding strategy looks smooth with significant reduction of streak artifacts. However, pixels located close to boundaries are not correctly restored with this relatively small number of iterations. However, by using the adaptive thresholding approximation with small $\beta = 0.001$, these drawbacks can be avoided.

In another study, we have used a star-shaped uniform reference image, shown in Fig (4)(a), such that a strong mismatch between the reconstructed image and the reference image is exist. We have used the dynamic reference approach given by Eq. (7) and the reference image is update side by side with the reconstructed image. We have used projection data measured over 32 projection views and a 3×3 window size. It is noticed from results, shown in Fig. 4, that as long as iteration proceeds, the dynamic reference images is corrected to the one that is used in Fig. 3(b).

IV. Conclusion

In this paper we discuss iterative reconstruction method that include *a priori* knowledge of reference image into account. In the framework of compressed sensing, the developed method take the form of two main steps. The first step is image update

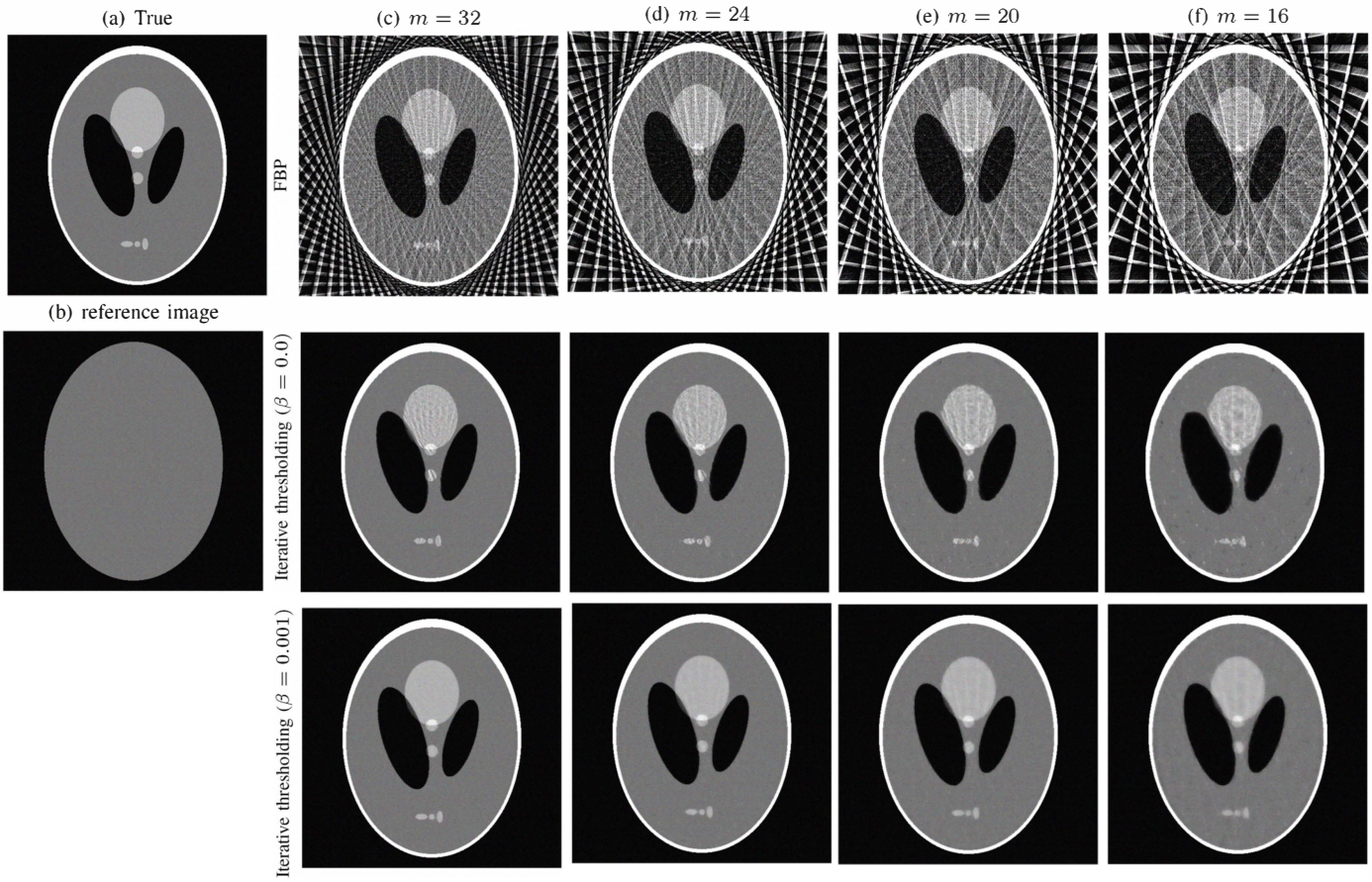

Fig. 3. Dynamic reference image, (a) is the current image estimate, (b) is the current reference image. The reference image for next iteration is computed using 3×3 window as in (c).

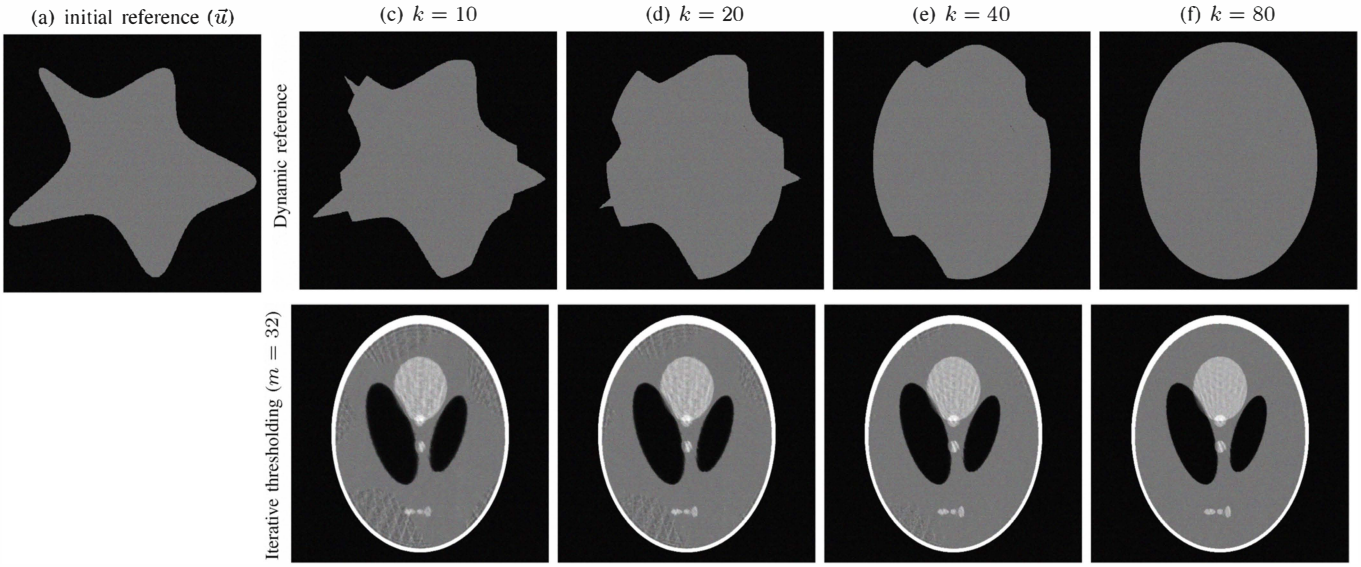

Fig. 4. Dynamic reference image, (a) is the current image estimate, (b) is the current reference image. The reference image for next iteration is computed using 3×3 window as in (c).

followed by a thresholding function. We develop adaptive thresholding function that can handle incorrect thresholding error. Moreover, we present a new approach toward control the misalignment between the reconstructed image and the *a priori* known reference image. The reference image is dynamically updated based on information from the current image estimate and the reference image in the previous iteration. Results presented in the paper is a preliminary results towards a full dynamic iterative image reconstruction protocol.

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